

Pipeline with Logic-Syntax-Driven Diagram Planning for Trustworthy LLM-Based Presentation Generation

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Abstract—Large Language Models (LLMs) have shown considerable potential for automating complex content generation tasks. However, there are significant risks related to their unconstrained use in educational contexts, including incorrect or incomplete explanations, illogical structures, and redundant content that compromise academic relevance. Addressing these risks requires not only automation but verifiable automation, meaning pipelines in which each generation step is grounded, validated, and structurally controlled. This paper introduces a retrieval-grounded pipeline for academic slide deck generation that prioritises trustworthy output through a top-down hierarchical verification at each stage. LLM outputs are constrained by structured prompts and verified against retrieved source evidence, increasing the likelihood that generated content is accurate, complete, coherent, and non-repetitive. This consists of outline structure, evidence retrieval, detailed scenario generation, diagram synthesis, and voiceover generation. A key contribution is diagram concept planning, in which slide content is mapped to a Prolog intermediate representation before D2 code generation. Semantically incorrect diagrams are less likely when using compact and structured symbolic representation, which highlights important conceptual relationships. Results show that this approach produces higher-quality diagrams across all evaluated metrics, confirming the benefit of interpretable intermediate representations in generation pipelines that combine symbolic and neural inference.

Index Terms—Retrieval-augmented generation, diagram generation, Prolog knowledge base, large language models, trustworthy AI

I. INTRODUCTION

LLMs have become known as useful tools for tasks requiring a large amount of knowledge, particularly when they can rely on external sources instead of generating answers exclusively from their internal parameters [1], [2]. This is especially important in generating educational content, where generating fluent text is insufficient. Additionally, the response must maintain a consistent structure to support the explanation with coherent illustrations and a logical flow. Under these constraints, grounding in retrieved sources becomes crucial. [3], [4].

Automatic presentation generation has improved significantly in recent years. Subsequent systems incorporated audience-aware planning and multi-stage generation, demonstrating that handling outline generation, slide writing, and visual design as distinct phases leads to better presentations [5], [6]. By using a workflow that is closer to

how people actually edit slides and by evaluating the results from multiple perspectives, such as content, design, and coherence, PPTAgent improved this concept [7]. Overall, these works suggest that slide generation works better as a structured pipeline than as a single prompt.

Nevertheless, one of the weakest aspects of many presentation systems is still automatic diagram generation. Despite the capabilities of state-of-the-art AI models in generative tasks, prompts generated automatically frequently lack critical specificity, resulting in diagrams that are generic or only weakly connected to key information. In educational presentations, where diagrams serve as essential tools for explaining relationships, processes, and hierarchies, this limitation becomes a major drawback.

Recent work on text-to-diagram generation shows that planning the diagram before rendering leads to better structure and easier editing [8], [9]. However, recent research shows that Retrieval Augmented Generation (RAG) by itself is insufficient to eliminate weakly grounded content [2]. A promising direction is neuro-symbolic reasoning. By converting natural language into symbolic representations before executing the next step of reasoning, systems such as Logic-LM, LINC, and SymbCoT enhance trustworthiness and control [10], [11], [12]. The results demonstrate that LLMs can extract symbolic descriptive representations. Unlike these works, we use symbolic structures to guide LLM inference in downstream tasks such as diagram generation to offer a solid ground for D2 diagram code generation and fast verification without image-to-text conversion.

Based on this idea, this paper proposes a RAG-based pipeline for educational slides' generation in which outline planning, per-slide scenarios, bullet writing, audio narration, diagram concepts, and D2 diagram code are produced in separate but connected stages. We explore whether diagram planning becomes more reliable when it is guided by an intermediate Prolog knowledge base, instead of relying only on free-form slide text. To study this, we compare three diagram concept generation strategies: direct diagram planning from RAG-grounded slide text, syntactic-based Prolog extraction followed by diagram planning, and LLM-based Prolog extraction followed by diagram planning.

The full pipeline incorporates embedding-based and Nat-

ural Language Inference (NLI)-based checks that compare the generated outputs with the retrieved evidence and other presentation modalities. This evaluation captures grounding, hallucination risk, diversity, and cross-modal consistency, resulting in a more trustworthy generation process. The main contributions of this paper are as follows:

- a multi-stage pipeline for educational slide generation that remains grounded in source material and preserves intermediate artefacts for inspection;
- the use of a compact symbolic intermediate representation based on Prolog facts and rules to define diagram-concept generation strategies that enhance semantic clarity;
- an evaluation framework that combines embedding-based grounding, NLI-based support and hallucination checks, diagram-text coherence, structural quality, diversity, and consistency across modalities.

The paper is organized as follows: Section II presents the framework and evaluation protocol; Section III discusses the results; and Section IV concludes with future directions.

II. PROPOSED FRAMEWORK

The system is designed to support educators with a simple and trustworthy generation of academic presentations. The input consists of a short topic prompt and an educational corpus (textbooks, lecture notes, etc.) organised as a database of embeddings for efficient retrieval. The system follows a top-down design in which outline planning, retrieval, slide-level scenario generation, diagram synthesis, and final rendering are executed in separate but interconnected stages. This section presents the proposed flow and the three diagram-concept generation strategies. In addition, the LLM-as-judge evaluation also uses `claude-haiku-4-5`. Retrieval, diagram rendering, and final export remain unchanged across settings, so the comparison isolates the effect of the generation backend.

A. System Architecture

The proposed pipeline runs in five successive stages (Fig. 1). Each stage creates explicit artefacts, such as outline files, retrieval results, scenario plans, generated presentation content, and rendered diagrams. These intermediate outputs can be inspected individually and reused in later runs, which allows for modular debugging and controlled experimentation. Planning decisions are kept apart from final wording decisions, which is a key design option of the framework. The system first decides what each slide should explain and how the diagram should be structured and only afterwards generates the final textual phrasing and rendered output.

1) *Outline Planning*: In the first stage, the system generates a structured slide outline from the input topic (TP) using a prompted LLM call. This outline, denoted PP , specifies the number of slides, their types, titles, and learning objectives, indicating the high-level framework of the presentation. This step acts as the global plan for the deck and determines the local goals that later stages must follow.

2) *Per-Slide Retrieval*: The system executes semantic retrieval at the slide level to obtain a concept for each slide. Retrieval is performed independently for each slide rather than just once for the entire topic to ensure that the next

generation phases are still linked to local evidence rather than just the global topic. The topic, slide title, and learning purpose are used to create a slide-specific query, which is matched against an indexed educational corpus. Each slide scenario s_i is then derived from the highest-ranked chunks.

3) *Per-Slide Planning*: A detailed per-slide scenario plan is generated through a sequence of LLM calls. First, the system produces a slide-level scenario that includes the planned bullet content, the teaching goal, a structural explanation plan, a speech plan, the diagram theme, and initial layout-related information. The layout includes a single diagram per slide, positioned either at the bottom or on the right side.

The diagram theme is mapped to a detailed description, d_i , which specifies all blocks and their interconnections. This description is then converted into D2 code, encoding all the information required for graphical generation. For the transformation to d_i , we considered three variants, all sharing the same slide description, s_i , and the same diagram theme, dt_i . In V1, a prompt generates d_i directly from s_i and dt_i . In V2 and V3, the information is first structured into an intermediate Prolog knowledge base, p_i . For V2, the base of facts and rules is automatically constructed using `spaCy`-based lexical and syntactic analysis. On the other hand, V3 extracts p_i through an additional LLM call. Thus, the outline, retrieved evidence, and slide scenario are identical across V1, V2, and V3, and the comparison isolates only the effect of the intermediate representation used for diagram planning.

The resulting D2 code is rendered into SVG and PNG assets. The pipeline separates semantic diagram planning from visual realisation: the prompt used for generating d_i specifies the intended structure of the diagram, while the prompt used for converting it into D2 code focuses on producing the concrete graphical assets inserted into the presentation. This separation improves both the conceptual clarity and the graphical quality of the resulting diagrams.

4) *Per-Slide Rendering*: The text is generated from the slide scenario and grounded in the retrieved evidence, while the rendered diagram assets produced from D2 codes are inserted into their slide positions together with the layout instructions derived from the scenario stage. The resulting deck is exported to PPTX through Marp, which is used here as a Markdown-to-PPTX backend. This choice provides a reproducible text-based rendering workflow, supports selective re-rendering of individual slides, and integrates with the intermediate JSON and diagram assets produced by the earlier stages. Compared with direct binary slide generation, this design makes the final deck easier to verify and modify, although it offers less low-level layout control than fully manual slide authoring. The pipeline also supports an optional video stage, in which synthetic speech is synchronized with slide frames to produce an MP4 presentation.

5) *Verification*: This pipeline further supports a thorough verification of all visual elements, assessing their grounding and diversity to ensure a trustworthy generation process. The step-by-step generation flow also makes the system easier to inspect than a one-shot prompting setup. The generated text, diagrams, and rendered outputs are thus compared against both the retrieved content and the different

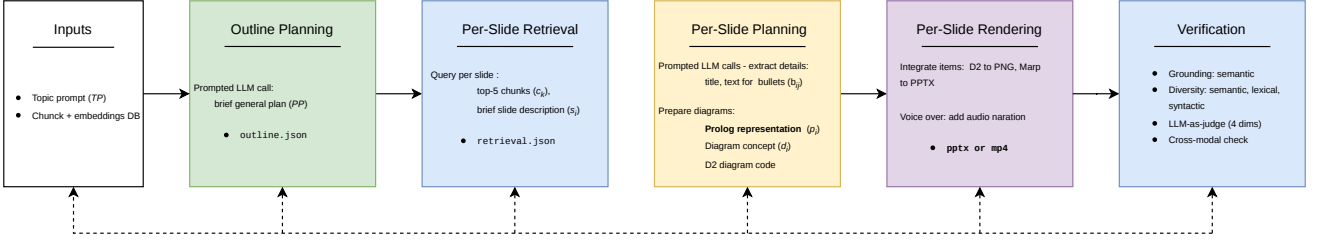


Fig. 1. Overview of the proposed retrieval-grounded pipeline for slide generation. The pipeline ensures a top-down, step-by-step generation and verification of all visual items. TP denotes the topic prompt, PP the presentation plan, c_k the retrieved evidence chunks, s_i the slide scenario, d_i the diagram description, and p_i the Prolog knowledge base. Dashed arrows indicate the backward verification checks applied to saved artefacts from each stage.

TABLE I
COMPARISON OF DIAGRAM CONCEPT GENERATION STRATEGIES

Item	V1	V2	V3
Intermediate Prolog representation	None	Extracted with spacy	Extracted with LLM
LLM calls per slide	3	3	4
Relation type	Implicit	Syntactic	Semantic
Rule generation	No	Pattern-based	Yes

presentation modalities in several downstream evaluations, with the retrieved evidence being reused across stages.

B. Diagram Generation

To enable a logical content of the diagrams, the slide semantics are converted into a structured diagram description, d_i that specifies the nodes, arrows, and layout. Then, a prompted LLM call converts it into executable D2 code. In this representation, nodes, edges, labels, and layout constraints are expressed explicitly in a textual graph format, for example through relations such as `input -> weighted_sum -> activation -> output`. D2 was chosen because it provides a deterministic and verifiable representation of diagrams. Compared with direct image generation or less structured visual generation methods, D2 makes it easier to preserve semantic relations.

Mathematical expressions are embedded directly in node labels as inline Unicode notation rather than typeset through a separate equation engine: symbols such as ∂ , $\sum$, $\prod$, and sub/superscripted terms (e.g., $\partial L / \partial w$, $\sigma(z_i)$) are written as plain text in the label fields of the d_i and carried verbatim into the corresponding D2 labels, where long expressions are wrapped at natural break points (e.g., at = or $\cdot$). Storing formulas as ordinary label text, rather than as rendered images, keeps them directly verifiable against the s_i .

We adopted Marp as the rendering backend because it offers a lightweight and easily maintainable workflow. The entire deck can be regenerated deterministically from a single Markdown source, with consistent styling enforced through theme files. Its native support for text-based diagram specifications, such as Mermaid, also simplifies integration and automatic re-rendering. While `python-pptx` remains more flexible, Marp provides a faster and more easily verifiable workflow for iterative slide development.

For generating the diagram concepts, d_i , we compare three approaches, which are summarized in Table I.

Variant V1: Direct diagram planning from RAG-grounded slide text. In the baseline setting, the diagram concept is generated directly from the grounded slide context. After receiving the s_i and the dt_i , the LLM generates a structured

description of the diagram. The model decides implicitly which concepts should become nodes and which relations should become arrows.

Variant V2: Diagram planning using a spacy-based Prolog representation. In the second case, the grounded s_i , concatenated with the dt_i , is first processed by a dependency-based extractor implemented with spacy. Subject-verb-object patterns are mapped to facts of the form `predicate(subject, object)`, while simple conditional structures are converted into rules of the form `conclusion(X, Y) :- condition(X, Y)`. This step produces a deterministic symbolic representation of the slide semantics before generating the detailed d_i . The extracted facts and rules are then injected into a prompt, where the LLM is instructed to map Prolog predicates to diagram relations and Prolog arguments to diagram nodes.

Compared with direct planning, this variant makes the relational structure explicit before the d_i is generated. As a result, the produced diagram is guided not only by the slide text but also by an intermediate symbolic representation that exposes entities and dependencies more clearly. Importantly, this variant does not change the upstream slide planning context. It uses the same s_i and retrieved evidence as V1 but restructures this information before diagram planning.

Besides supporting diagram generation, this variant also enables a simpler verification process. Since nodes and arrows can be compared directly against the extracted Prolog facts and rules, the resulting diagram can be partially checked without relying on image-to-text recovery. This makes the verification more interpretable and computationally lighter. However, because the representation is derived from dependency patterns rather than deeper semantic interpretation, it may miss implicit relations or produce incomplete structures when the slide text contains compressed technical phrasing.

Variant V3: Diagram planning using LLM-based Prolog representation. An extra LLM call extracts a Prolog knowledge base using the grounded slide text. In the same manner as V2, these symbolic elements are then embedded into the diagram concept prompt. This method represents an additional model-generated intermediate representation while recovering relations that are difficult to capture by syntactic patterns when compared to V2. Similar to V2, Prolog-based representation can assist the verification of the generated D2 code without using image-to-text models.

In all variants, the diagram planning step provides a structured description that specifies nodes with labels and shapes, directed arrows between nodes, a layout direction, and

optional layout notes. This description is then passed to the same D2 code generation stage, so the comparison focuses on the impact of the intermediate representation used prior to diagram rendering. The pipeline has downstream quality checks that examine alignment between generated content and retrieved evidence, diagram relevance, and cross-modal consistency across text, visual items, and narration.

C. Evaluation Protocol

For a comprehensive evaluation of the generated presentations, we use metrics that capture similarity with retrieved evidence, lexical and cross-variant diversity, and slide-level quality assessed by an LLM judge. This verification is designed to detect inconsistencies across all visual items included in the presentation, identify problematic components, and prevent common issues such as hallucinations, illogical content, and unwanted repetition of ideas. By leveraging the step-by-step generation flow, the verification stage can relate items across multiple levels of the hierarchy, ensuring both accurate grounding and sufficient diversity.

1) *Grounding*: Grounding for an item is assessed by computing the cosine similarity between its embedding and the embeddings of all connected items produced in earlier steps. All embeddings are generated with `all-MiniLM-L6-v2` [13]. For example, the grounding of a bullet b_{ij} on the i^{th} slide can be evaluated against all prior items, namely the s_i , the PP , the TP , and the top-K retrieved corpus chunks, c_k ($k = 1, \dots, K$). This backward chain enables fine-grained verification and contributes to improved explainability. However, cosine similarity with the original chunks does not fully guard against hallucination and could be complemented by graph-based metrics for a refined consistency check.

Grounding can be assessed either individually (i.e., for a single bullet, diagram, or s_i) or at the group level. In the latter case, all similar items produced within a step are concatenated and compared with the corresponding aggregated items from previous steps, e.g., the concatenation of all bullets on slide i can be evaluated against s_i , PP or TP ; similarly, the concatenation of all bullets across the entire presentation can be compared with PP or TP . The Prolog bases offer important advantages for diagram verification, as they can follow the same verification flow. Given the low error rate in generating diagrams from these representations, their use enables more reliable diagram evaluation even without relying on image-to-text conversion, which is significantly more resource-intensive.

Some key metrics are exemplified. We considered the mean grounding of individual bullets, as well as the grounding of per-slide bullet lists across the presentation, computed with respect to the corresponding s_i :

$$G_{s-b} = \frac{1}{N_b} \sum_{i,j} \frac{\mathbf{e}(b_{ij})^\top \cdot \mathbf{e}(s_i)}{\|\mathbf{e}(b_{ij})\| \cdot \|\mathbf{e}(s_i)\|}, \quad (1)$$

$$G_{s-b_{all}} = \frac{1}{N_S} \sum_i \frac{\mathbf{e}(B_i)^\top \cdot \mathbf{e}(s_i)}{\|\mathbf{e}(B_i)\| \cdot \|\mathbf{e}(s_i)\|}, \quad (2)$$

where N_b is the total number of bullets in the presentation, N_S is the number of slides and $B_i = \text{concat}_j(b_{ij})$ is the bullet list from the slide i . In the same manner as in

equation (2), we define the mean grounding per diagram by comparing d_i and s_i , and the mean grounding per Prolog representations by comparing p_i with s_i . These metrics are particularly important because LLMs usually fail at generating clear and correct details, even though they are generally reliable at outlining high-level plans.

In addition to the embedding-based similarities above, we also report NLI-based support against the retrieved chunk set of each slide, denoted $C_i = \{c_{ik}\}_{k=1}^{K_i}$. For a generated text x , using $g(x, C_i) = \max_{c \in C_i} P_{\text{ent}}(c, x)$, we define:

$$h(x, C_i) = \begin{cases} 1, & \text{if } \max_{c \in C_i} P_{\text{con}}(c, x) > g(x, C_i) \\ & \text{or } g(x, C_i) < \tau \\ 0, & \text{otherwise.} \end{cases} \quad (3)$$

where P_{ent} and P_{con} denote the entailment and contradiction probabilities returned by the NLI model, and τ is a support threshold. We used `DeBERTa-v3-large` as NLI backbone and $\tau = 0.5$.

The corresponding factual precision indicates the proportion of non-hallucinated items. For individual bullets and concatenated bullet lists, respectively, it is computed as follows:

$$FP_{s-b} = 1 - \frac{1}{N_b} \sum_{i,j} h(b_{ij}, C_i),$$

$$FP_{s-b_{all}} = 1 - \frac{1}{N_S} \sum_i h(B_i, C_i). \quad (4)$$

Thus, FP_{s-b} measures how often individual bullets are supported by the retrieved evidence, whereas $FP_{s-b_{all}}$ performs the same check for the full bullet list of each slide.

For diagrams, each directed edge extracted from d_i is verbalized as a canonical relation sentence, e.g., “A leads to B.” If $R_i = \{r_{im}\}_{m=1}^{M_i}$ denotes the set of such arrow-derived relations for slide i , we compute:

$$FP_{s-d} = 1 - \frac{1}{N_R} \sum_{i,m} h(r_{im}, C_i), \quad (5)$$

where $N_R = \sum_i M_i$. Thus, FP_{s-d} captures the rate of non-hallucinated relations. Additionally, diagram concepts d_i are verified against slide scenarios s_i :

$$G_{s-d} = \frac{1}{N_S} \sum_i \frac{\mathbf{e}(d_i)^\top \cdot \mathbf{e}(s_i)}{\|\mathbf{e}(d_i)\| \cdot \|\mathbf{e}(s_i)\|}. \quad (6)$$

2) *Diversity*: Diversity assessment is important from two perspectives. Cross-similarity between bullet texts or diagram concepts within a presentation helps detect undesired repetitions, while cross-similarity across presentations generated for similar topic prompts helps evaluate the creativity supported by the pipeline. Diversity is assessed at the lexical, syntactic, and semantic levels. For cross-similarity in a presentation, we compared an item with all the other items from its category, e.g., $D_{b_{ij}}$ indicates the mean diversity of the bullet text b_{ij} against the other bullets, and D_{d_i} offers a similar evaluation for the diagram concept d_i :

$$D_{b_{ij}} = \sum_{m \neq i, n \neq j} \frac{dd(b_{ij}, b_{mn})}{N_b}, D_{d_i} = \sum_{m \neq i} \frac{dd(d_i, d_m)}{N_S} \quad (7)$$

TABLE II
MEAN GROUNDING AND PAIRED ITEMS DIVERSITY OVER THE SLIDES

V.	G_{s-b}	H_{s-b}	$G_{s-b_{all}}$	$H_{s-b_{all}}$	G_{s-d}	H_{s-d}	G_p	D_{B_i}		D_{D_i}	
								D-2	TED	D-2	TED
V1	0.458	0.938	0.578	0.906	0.407	0.941	0.404	0.852	0.550	0.913	0.574
V2	0.458	0.938	0.578	0.906	0.407	0.941	0.404	0.852	0.550	0.913	0.574
V3	0.458	0.938	0.578	0.906	0.407	0.941	0.404	0.852	0.550	0.913	0.574

TABLE III
DIAGRAM CONCEPT AND PROLOG REPRESENTATION EVALUATION

Topic	Var.	G_{p-s}	CC	FA	LR	DR	HU-A
T1	V1	—	0.743	0.807	0.869	0.778	4.0
	V2	0.413	0.753	0.808	0.863	0.778	4.5
	V3	0.427	0.756	0.794	0.872	0.783	4.75
T2	V1	—	0.855	0.844	0.784	0.889	4.17
	V2	0.395	0.837	0.835	0.789	0.873	4.58
	V3	0.403	0.859	0.873	0.781	0.892	4.83
Mean	V1	—	0.799±0.079	0.826±0.026	0.827±0.060	0.834±0.078	4.04
	V2	0.404±0.013	0.795±0.059	0.822±0.019	0.826±0.052	0.826±0.067	4.54
	V3	0.415±0.017	0.808±0.073	0.834±0.056	0.827±0.064	0.838±0.077	4.79

Here, dd denotes the adopted diversity distance. For lexical diversity, we used Distinct-2 [14], which measures the ratio of unique 2-grams to the total number of tokens. We also used self-BLEU, where each item is treated as a hypothesis and scored against the others. Mean Tree Edit Distance (TED) was used to assess syntactic diversity by comparing dependency-parse structures. Finally, the cosine similarity between embeddings complements this analysis with a semantic perspective. All metrics were scaled such that higher values correspond to more diverse items. To compare a presentation against another, we used the full list of bullets and the full list of diagram concepts, respectively.

3) *Overall quality assessed by humans and an LLM judge*: Following [15], a separate LLM call scores each d_i on four dimensions in $[0, 1]$: concept coverage (CC), factual accuracy (FA), lack of repetition (LR), and diagram relevance (DR). The overall score is their mean. Human assessment (HU-A) averages ratings for educational relevance and clarity from 3 students and 3 teachers (1–5; 5 = best).

III. RESULTS AND DISCUSSION

We evaluate the generated presentations on two topics: *Maxwell's Equations and Electromagnetic Waves* (T1) and *Bayes' Theorem* (T2). The retrieval corpus consists of text chunks extracted from freely available course materials (<https://ocw.mit.edu/>): 1163 chunks from the 6.013 *Electromagnetics and Applications* course and 1576 chunks from *An Introduction to Machine Learning* and the 6.867 *Machine Learning*. All materials consist predominantly of textual explanations. During preprocessing, images were converted to text with Claude Haiku 4.5, while formulas were transcribed to LaTeX with Claude Sonnet 4.5. For each topic, we perform six independent runs and generate one presentation with each diagram-planning variant (V1, V2, and V3), resulting in 36 presentations.

Within each topic-run setting, the same PP , c_k , and grounded s_i are reused across V1, V2, and V3, so that only the diagram-planning strategy varies. Each presentation contains 8 slides, although the framework supports variable deck lengths. For consistency, all reported metrics are scaled so that higher values indicate better results. We first analyse grounding and diversity within each presentation to assess the effect of the three diagram-planning variants and then we examine diversity across repeated runs for the same topic to evaluate the creativity supported by the pipeline.

TABLE IV
DIAGRAM CONCEPT EVALUATION: CROSS-DIVERSITY ACROSS DIAGRAM-PLANNING VARIANTS FOR THE SAME TOPIC

Topic	Distinct-2			1 – BLEU	TED	1 – Cosine
	V1	V2	V3			
T1	0.827	0.841	0.831	0.585	0.698	0.120
T2	0.836	0.848	0.824	0.737	0.713	0.187
Mean	0.831±0.006	0.845±0.005	0.827±0.005	0.661±0.107	0.706±0.011	0.153±0.047

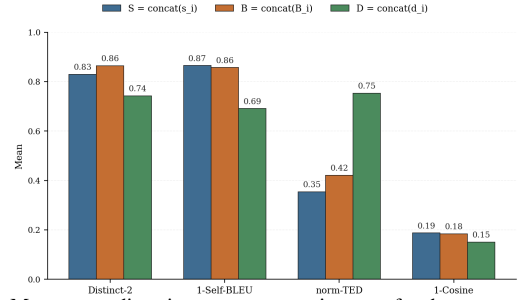


Fig. 2. Mean cross-diversity across successive runs for the same topic. We compare the concatenated slide scenarios $S = \text{concat}_i(s_i)$, concatenated bullet sets $B = \text{concat}_i(B_i)$, and concatenated diagram descriptions $D = \text{concat}_i(d_i)$ using Distinct-2, 1-Self-BLEU, norm-TED, and 1-Cosine.

A. Quantitative analysis

1) *Grounding and diversity across the slides*: Grounding of bullet texts and diagrams is assessed individually, using the overall mean of G_{s-b} and G_{s-d} defined according to (2). Furthermore, we evaluated the grounding of the per-slide bullet list, using $G_{s-b_{all}}$ defined in (3).

The results in Table II show that the generated bullets and diagram concepts remain well grounded in the slide scenarios. Jointly evaluating the full bullet list yields stronger grounding than scoring bullets individually, while V3 achieves the best diagram grounding and Prolog-to-scenario grounding. The hallucination-oriented scores further indicate that the generated content remains faithful to the c_k . Overall, the results confirm high diversity and a low risk of repetitions.

Additional results related to the diagrams are presented in Table III. These tests focused on the impact of diagram planning strategies (V1, V2 and V3), using an LLM as judge for evaluating the diagram concept. Additionally, for V2 and V3, we evaluated the grounding score G_{p-s} , which measures the mean cosine similarity between the p_i and s_i . As shown in Table III, no single variant dominates all evaluation criteria; however, V3 achieves the highest mean results in grounding, concept coverage, factual accuracy, and diagram relevance, while logical relevance remains comparable across variants. V3 also obtains the highest mean HU-A score (4.79).

2) *Diversity of presentations related to similar topics*: Fig. 2 reports the mean cross-diversity across successive runs for the same topic, computed over the concatenated slide scenarios S_i , bullet sets B_{ij} , and diagram descriptions D_i . The results show consistent variation across repeated generations, indicating that the pipeline does not simply reproduce near-identical presentations under different runs. Furthermore, Table IV shows that, even when V1, V2, and V3 are applied to the same topic and slide scenarios, they do not collapse to identical diagram concepts. The generated descriptions remain distinct across variants, with V2 achieving the highest Distinct-2 score, while 1-BLEU, TED, and 1-Cosine also confirm variation across variants.

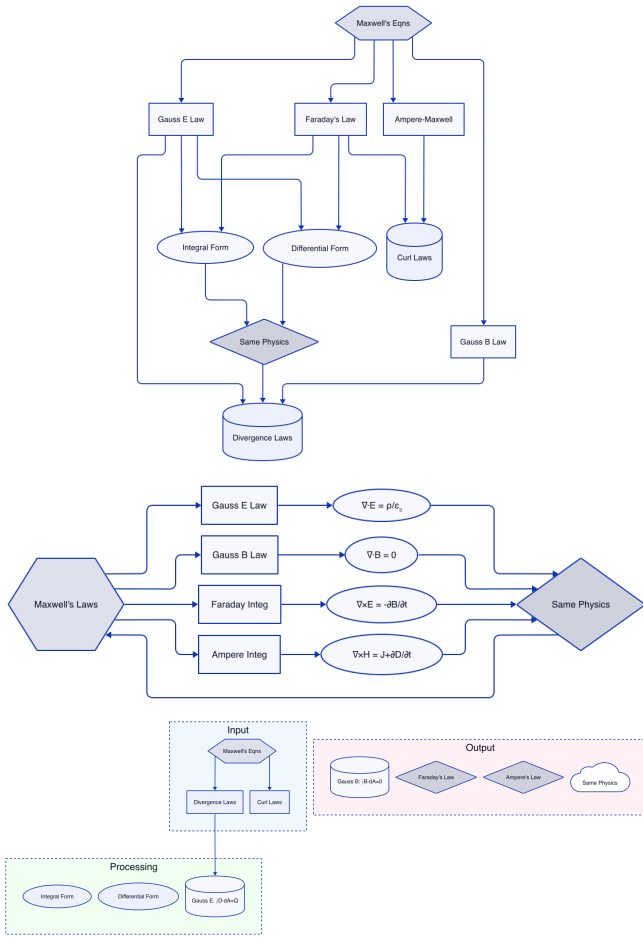


Fig. 3. Diagrams for the slide *The Four Maxwell's Equations* [Maxwell's Equations and Electromagnetic Waves topic]: V1 (top); V2 (middle); V3 (bottom).

B. Qualitative Analysis

Fig. 3 compares the diagrams produced by the three planning variants for the slide *The Four Maxwell's Equations*. V1 (left) captures the topic only at a generic conceptual level without explicitly representing the equations themselves. V2 (middle) is more informative, as it directly instantiates the differential-form Maxwell equations in the diagram nodes, making the structure more faithful to the source content and more useful instructionally. V3 (right) preserves a clear staged organization but includes fewer explicit mathematical relations than V2. This comparison suggests that the symbolic intermediate structure improves diagram grounding by preserving equation-level semantics.

IV. CONCLUSIONS AND FUTURE WORK

This paper introduces an RAG-based pipeline for automated educational slide generation. The proposed framework is modular and enables verification of all visual and text items inserted in the presentation. Prolog-based representations are used to guide a trustworthy and explainable diagram generation. These representations are extracted either by syntactic analysis (V2) or by LLM (V3).

The experiments indicate the items generated for the presentations are diverse and well grounded in the topics. The symbolic intermediate representation improves diagram-level quality and provides a more interpretable basis for

verification. The Prolog knowledge base also enables reliable verification of the diagram plan without requiring image-to-text models. Future work will include evaluating the approach on a larger benchmark and refining the symbolic layer through hybrid symbolic-LLM methods.

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